Practice Session 5:

Working with PAAS (Azure App Service) – CICD Approach

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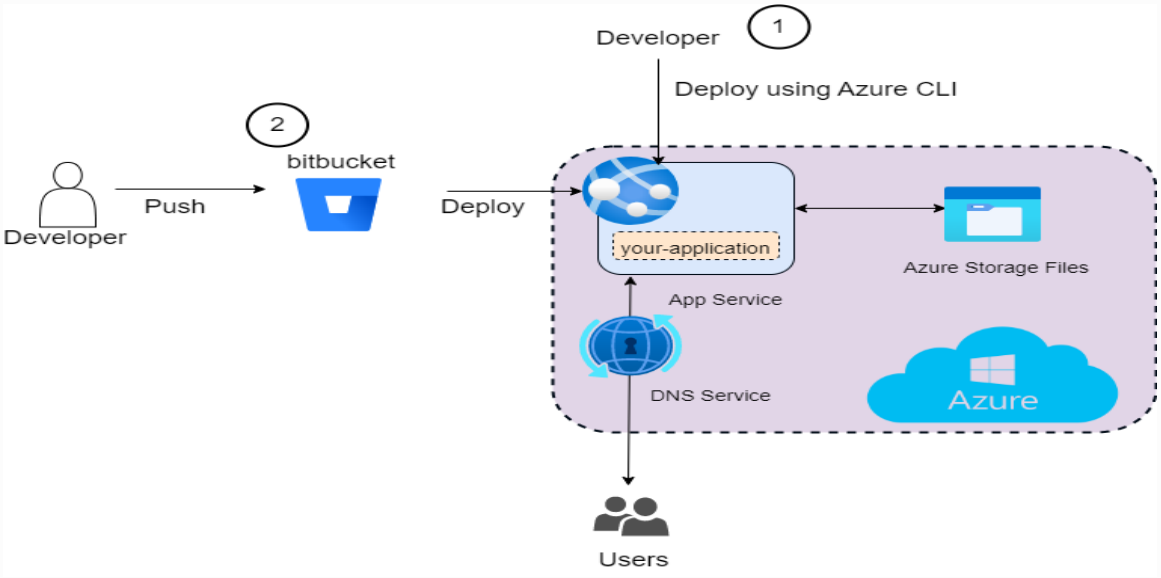
Practice 5 - Platform as a Service

During this practice session, we'll look into Microsoft Azure's Platform as a Service (PaaS). We'll focus on deploying web applications as PaaS offerings and examining various features of this cloud platform.

Applications can be deployed onto PaaS in Multiple ways:

Using CLI (Command Line Interface) - Manually deploying the code

1. Git-based (GitHub) - Linking code from your app repository
2. Using docker image registries - Linking a Docker image from a container registry



References

1. Azure Cloud documentation: https://docs.microsoft.com/en-us/azure/?product=featured
2. Azure App Service: https://learn.microsoft.com/en-us/azure/app-service/
3. Azure Storage: https://learn.microsoft.com/en-us/azure/storage/common/storage-introduction

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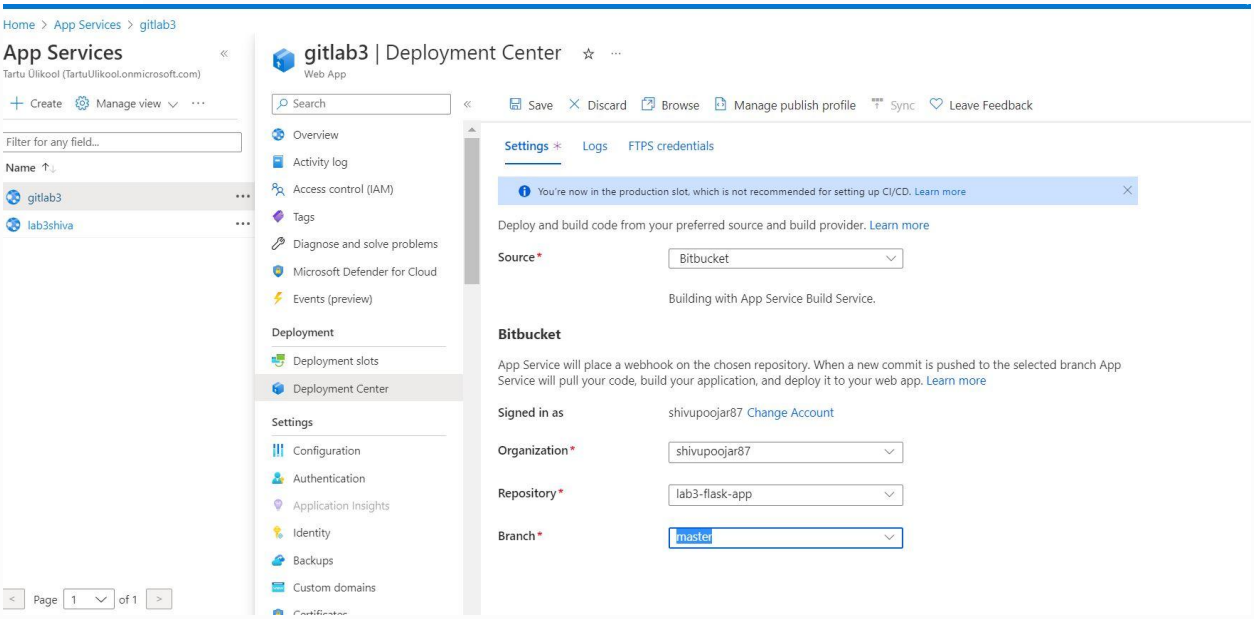
Exercise 5 Deploying python flask application in Azure using GitHub

In this exercise, you will learn to deploy your flask application code from the Bitbucket repository to Azure App Service. Here are the following tasks to be performed:

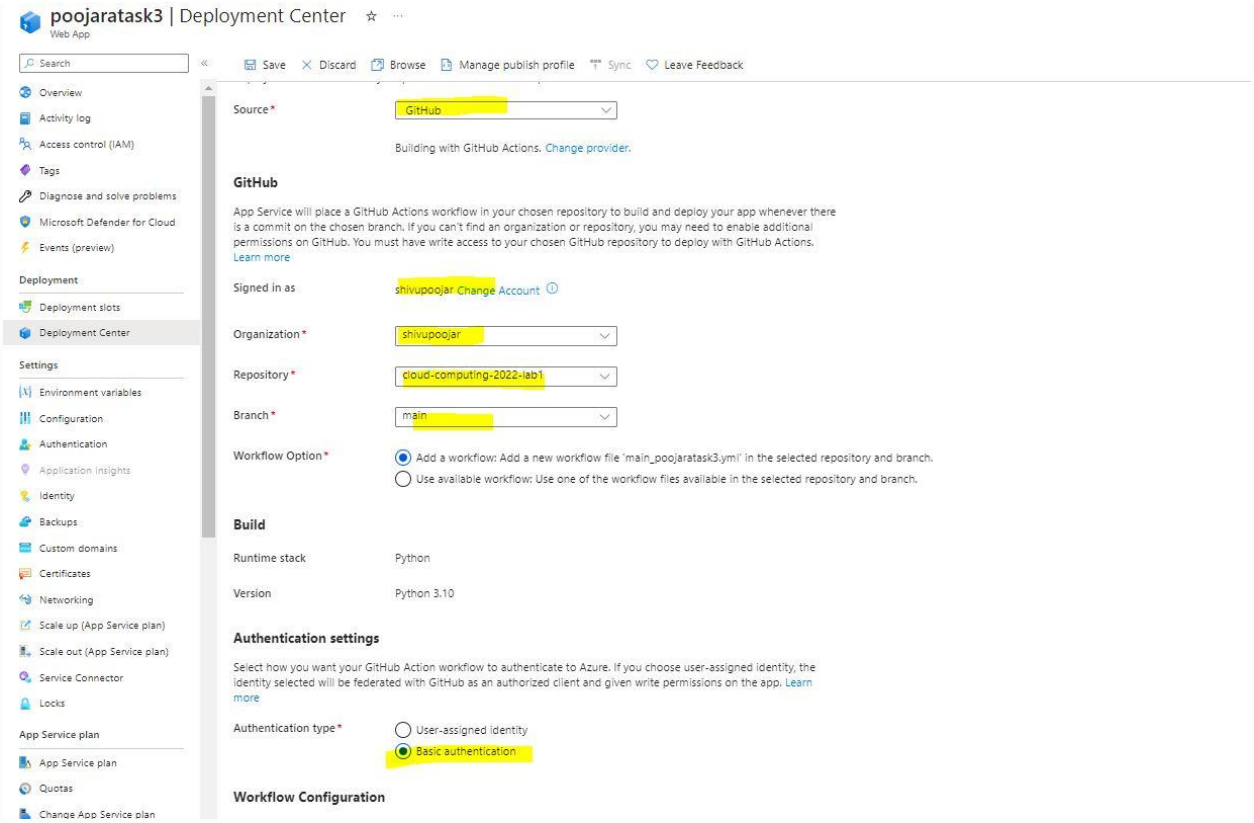
1) Integrate with Azure App Service with Bitbucket/GitHub code.
2) Check for the deployment status.
3) Modify the application, commit to Bitbucket/GitHub code
4) Finally, you should see your application running with modified code.

Task 1. Integrate Azure App Service with Bitbucket/GitHub code.

- Create a New App Service application
  - Resource group: `lab3`
  - Name: `FREELY CHOOSE`
  - Publish: `Code`
  - Runtime Stack: `Python 3.9`
  - Region: `North Europe`
  - Click on Review and Create
- Now, integrate with Bitbucket/GitHub, Go to **Deployment Center** in left panel and set the following values
  - For Bitbucket:
    - **Source**: Select `bitbucket`
    - **Organization** : Provide your bitbucket username
      - Provide permission to access by logging in to bitbucket.
    - **Repository**: Choose your repo `lab3app`
    - **Branch**: `master`
  - Finally, Save the configurations by Click on **Save** button at the top.

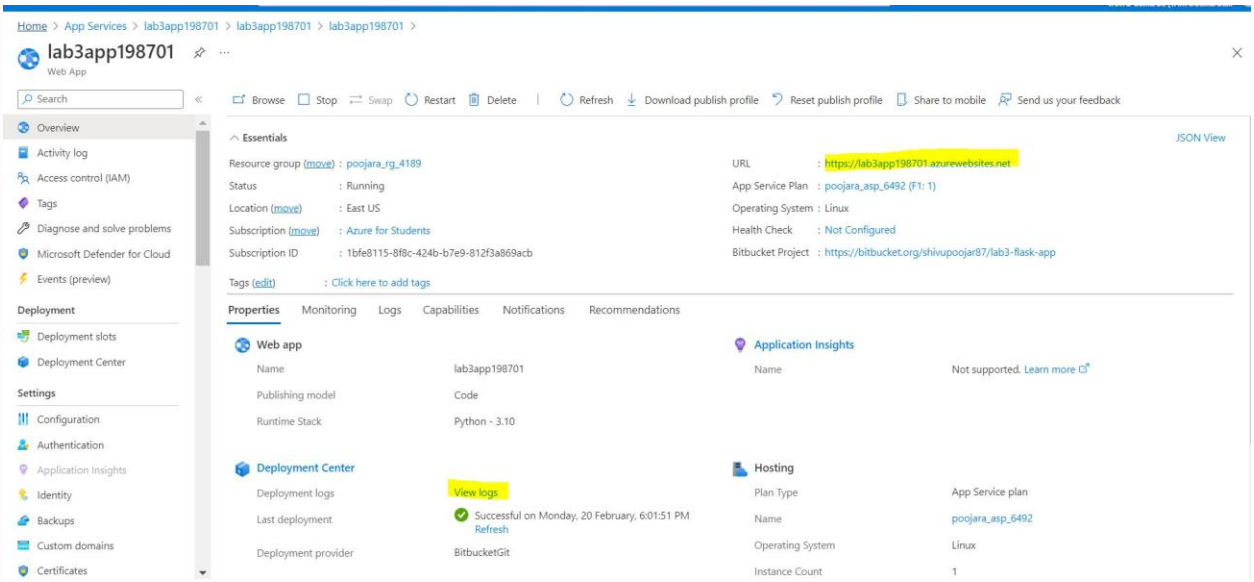


- For GitHub:
  - **Source**: Select `github`
  - **Sign in as**" Sign in to your GitHub account
    - Provide permission to access by logging in to GitHub.
  - **Organization** : Provide your GitHub username
  - **Repository**: Choose your repo `lab3app`
  - **Branch**: `main`
  - **Workflow Option**: Choose `Add Workflow...`
  - **Authentication type**: User/assigned identity
    - Have Azure generate new identity automatically
- Finally, Save the configurations by Click on **Save** button at the top.



Task 2: Check for the status of the deployment.

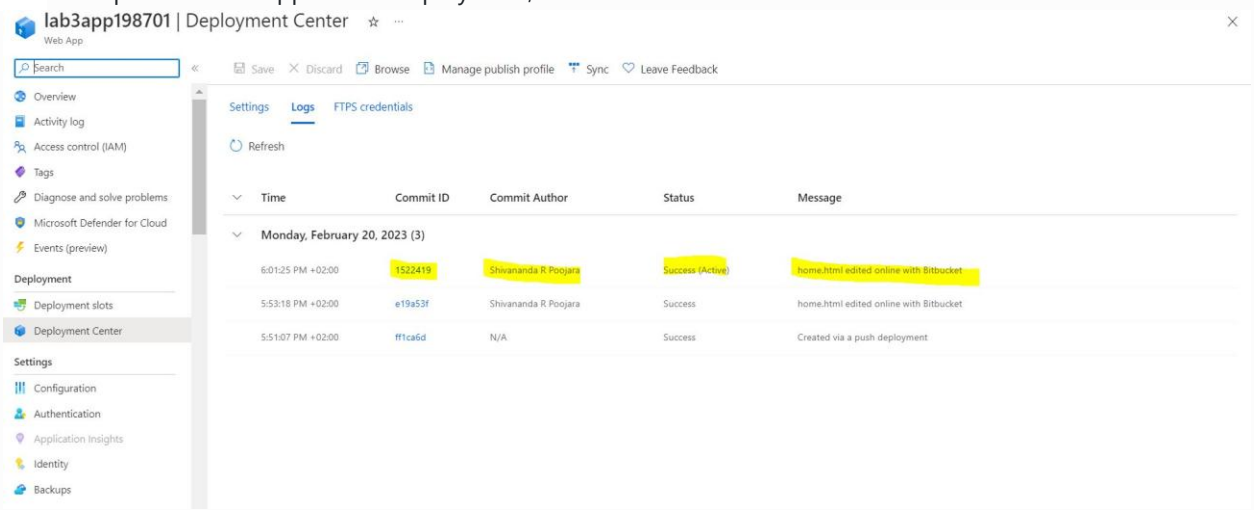
- Click on **View Logs** under Deployment Center



- Click on the URL of your project (In overview, as shown in the above figure.), and you should see your application running (It may take a few minutes to get up and running).

Task 4: Modify the application and commit to Bitbucket/GitHub code.

- Code can be committed using git commands from the system or using Bitbucket/GitHub web interface. However, to commit using git commands with your private repository, needed to setup Bitbucket ssh keys using this document or you can use the Bitbucket app password created in the step Ex 3.1 (PS!! Choice is yours)
- Edit the `home.html`, like Practice Session 1, to display your name on the web page (Make something interesting on your web page :)).
- Commit the code.
- Check the logs in App Service in **Deployment Center**. You should see the commit status with an update on the application deployment, as shown below.



- You should see your application running with the modified code.